Prompt:

Write a complete Python script that reads reservoir sensor from an Excel file named Data_01_Updated.xlsx (sheet name: 'Sensor Log', starting data at row index 3, using columns Timestamp and Depth). The script should perform numerical analysis, non-linear regression, and generate a standalone, beautifully styled 3-tab HTML dashboard named Lab02_acueza.html using Chart.js.-

Set OpenBLAS/MKL/OMP threading environment variables to 1 to avoid thread conflicts, load the sensor data, and establish a numeric time axis t with 0.25-hour steps.

1. Calculate the first derivative (dh/dt) and second derivative (d^2h/dt^2) using forward, backward, and central finite-difference approximations. Identify maximum filling rates and acceleration metrics.
2. Fit a 4-Parameter Logistic (4PL) regression model $h(t) = c + \frac{a}{1+e^{-k(t-t_0)}}$ to the depth data using scipy.optimize.curve_fit. Compute standard errors, residuals, Sum of Squared Errors (SSE), Total Sum of Squares (SST), Coefficient of Determination (R^2), and Residual Standard Error (s).
3. Calculate the area under the curve using both continuous integration (scipy.integrate.quad) on the fitted model and discrete trapezoidal integration (scipy.integrate.trapezoid) on the raw telemetry.
4. Calculate t-statistics and p-values for each of the 4 fitted parameters using Student's t-distribution and degrees of freedom.
5. **3-Tab HTML Dashboard Output:** Generate a fully responsive HTML file featuring:
   - **Tab 1 (Raw Data & Derivatives):** Displays three separate Chart.js line charts—the raw water depth observations, the first derivative (dh/dt), and the second derivative (d^2h/dt^2)—each accompanied by a professional explanation box
   - **Tab 2 (Fitted Logistic Curve):** Displays an overlay chart comparing raw sensor observations against the fitted 4PL curve, with an explanatory description.
   - **Tab 3 (Residuals & Diagnostics):** Displays structured HTML tables for Goodness-of-Fit & ANOVA metrics (including R^2, SST, SSE, and Residual Standard Error), Parameter Significance estimates (c, a, k, t_0 with standard errors, t-stats, and p-values), Volumetric Integration comparisons, and a scatter plot of model residuals with a clear explanatory box.
   - Use a cohesive purple/magenta professional color palette (#4a235a, #5b2c6f, #e75480) and clean CSS styling throughout.